St. John College of Engineering and Management
Autonomous Institute
(A Christian Religious Minority Institution)
Approved by AICTE and DTE, Affiliated to University of Mumbai / MSBTE
DTE Code : 3218 AICTE Permanent ID : 1-4790201
NAAC Accredited with Grade A+, Three Programs NBA Accredited

Internal Assessment Examination- I SET-B

Course Name – Engineering Physics **Semester- I**
Course Code- 25FEBS102 **Date- 18/10/25**
Maximum Marks- 10 **Time-12.00 -12.30 pm**
All Question are Compulsory
Assume Suitable data wherever required

	Question	Marks	CO	Level
Q1	**Attempt all Questions**			
a	**How are Miller indices determined?** a) By measuring the colour of the crystal b) By calculating the reciprocal of the intercepts of a plane with the crystal axes c) By weighing the crystal d) By determining the volume of the crystal	1	1	L1
b	**What is the effect of increasing temperature on the Fermi level in semiconductors?** a) Fermi level shifts toward the conduction band. b) Fermi level shifts toward the valence band. c) Fermi level remains constant. d) Fermi level moves toward the middle of the band gap.	1	2	L1
c	**What is spontaneous emission?** a) Emission of a photon without external influence b) Emission of a photon due to external radiation c) Absorption of a photon d) Refraction of light in a medium	1	5	L2
Q2. a	Define Space Lattice, Basis. What is a unit cell?	3	1	L3
	or			
Q2. b	Draw the planes for following Miller indices : (200), (110), (113)	3	1	L3
Q3. a	With the help of a diagram show the position of fermi level in a n-type semiconductor.	2	2	L2
	or			
Q3. b	Find the resistivity of intrinsic Ge at 250°K given the density of carriers as 3.0×10^{19} m^{-3}. The mobility of electrons and holes is 0.30 m^2/V-sec and 0.10 m^2/V-sec. respectively.	2	2	L2
Q4. a	Explain the term V number.	2	5	L2
	or			
Q4. b	Calculate the numerical aperture of a fibre with core index 1.46 and cladding index 1.40.	2	5	L2